

SAT Hard Question 9

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Question : Exponential Growth

Elizabeth has \$4500 in an account.

Each year she expects to have ****11% more**** money in the account than the previous year. **Which of the following models best describes how**

Elizabeth expects the money in her account to change over time?

- A. Decreasing exponential
- B. Decreasing linear
- C. Increasing exponential
- D. Increasing linear

Answer Explanation

Correct Answer: C. Increasing Exponential

Explanation:

- **A. Decreasing exponential:** Incorrect, because the amount is increasing.
- **B. Decreasing linear:** Incorrect, because the amount is increasing, and the growth is not linear.
- **C. Increasing exponential:** ****Correct!**** The money grows by a fixed percentage each year.
- **D. Increasing linear:** Incorrect, because the growth is not constant but rather proportional.

Key Concept: ****Exponential growth occurs when a quantity increases by a fixed percentage each period.****

Question : Air Temperature and Altitude

The function

$$T(x) = \frac{20,000 - 6.5x}{1,000}$$

gives the estimated air temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), at an altitude of x meters.

If the estimated temperature surrounding the hot air balloon is **15°C** , which of the following is closest to the altitude x of the hot air balloon?

- A. 20
- B. 769
- C. 1,333
- D. 5,385

Answer Explanation: Air Temperature

Correct Answer: B. 769

Solution:

$$15 = \frac{20,000 - 6.5x}{1,000}$$

Multiply both sides by 1,000:

$$15,000 = 20,000 - 6.5x$$

Solving for x :

$$x = \frac{20,000 - 15,000}{6.5} = \frac{5,000}{6.5} \approx 769.23$$

Thus, the closest answer is ****769****.

Question : Finding a Possible Value of $a + b$

The function f is defined as:

$$f(x) = (x - a)(x - b)$$

where a and b are integer constants.

Given that:

$$f(17) > 0, \quad f(20) < 0, \quad f(23) > 0$$

What is one possible value of $a + b$?

- A. 39
- B. 38
- C. 37
- D. 36

Answer Explanation: Finding $a + b$

Correct Answer: A. 39

Solution:

$$f(x) = (x - a)(x - b)$$

The function changes sign at $x = a$ and $x = b$, meaning $f(x)$ is negative between a and b .

Given:

$$f(17) > 0, \quad f(20) < 0, \quad f(23) > 0$$

This implies the roots are between 17 and 23, so we assume:

$$a < 17, \quad b > 20$$

Given $a + b$ is an answer choice, we check:

$$a + b = 39$$

Which satisfies the conditions.

Question: What Percent of b is a ?

The positive number a is 2,106% of the sum of the positive numbers b and c , and b is 81% of c .

What percent of b is a ?

Options:

- A. 4,706
- B. 2,600
- C. 38.12
- D. 21.87

Answer Explanation: What Percent of b is a ?

Correct Answer: 4,706%

Solution:

- Given $b = 0.81c$.
- Express a in terms of c :

$$a = 21.06(c + 0.81c) = 38.065c.$$

- Find $\frac{a}{b}$:

$$\frac{a}{b} = \frac{38.065c}{0.81c} = 47.06.$$

- Convert to percentage:

$$a = 4706\%b.$$

Final Answer: 4,706%.

Question: Identifying the Base or Coefficient in $f(x)$

For the function f , for every increase of 2 in the value of x , the value of $f(x)$ increases by a factor of c , where c is a constant.

Which of the following equivalent forms of function f displays the value of c as the base or the coefficient?

Options:

A. $f(x) = 42(2)^{6x}$

B. $f(x) = 42(8)^{2x}$

C. $f(x) = 42(64)^x$

D. $f(x) = 42(4096)^{\frac{x}{2}}$

Answer Explanation: Identifying the Base or Coefficient in $f(x)$

Correct Answer: $f(x) = 42(4096)^{\frac{x}{2}}$

Solution:

- Option A: $42(2)^{6x}$ - The base is 2^6 , which is not what we need.
- Option B: $42(8)^{2x}$ - The base is 8^2 , again incorrect.
- Option C: $42(64)^x$ - This is equivalent to 2^{6x} , not the required form.
- Option D: $42(4096)^{\frac{x}{2}}$ - This correctly expresses the factor c .

Final Answer: $42(4096)^{\frac{x}{2}}$.

Question: Validity of a Research Study

A study investigated the effect of magnesium supplementation on sleep quality. 190 adults were randomly selected, and their sleep efficiency was measured.

Participants were randomly assigned to take a magnesium supplement or a placebo for 6 weeks. The study concluded that magnesium improved sleep quality.

What feature of this study supports this conclusion?

- A. There were more than 100 participants in the study.
- B. The participants were selected at random from the community center.
- C. The sleep efficiency of each participant was measured again at the end of 6 weeks.
- D. Each participant was randomly assigned to take a magnesium supplement or a placebo.

Answer Explanation: Validity of a Research Study

Correct Answer: D.

Solution:

- ****Random assignment**** ensures that the two groups (magnesium vs. placebo) were similar at the start, meaning that any difference in sleep quality at the end can be attributed to the magnesium supplement.
- This eliminates confounding variables and makes it possible to conclude that magnesium caused the improvement.

Final Answer: D.

Question: Finding a Point on Line t

A circle in the xy -plane has its center at $(-1, 1)$. Line t is tangent to this circle at the point $(7, -6)$.

Which of the following points also lies on line t ?

- A. $(0, 7)$
- B. $(6, 9)$
- C. $(14, 2)$
- D. $(15, 1)$

Answer Explanation: Finding a Point on Line t

Correct Answer: C.(14, 2)

Solution:

- Compute the slope from the center $(-1, 1)$ to the tangent point $(7, -6)$:

$$\frac{-6 - 1}{7 - (-1)} = \frac{-7}{8}.$$

- The slope of line t is the negative reciprocal:

$$\frac{8}{7}.$$

- Equation of line t using $(7, -6)$:

$$y + 6 = \frac{8}{7}(x - 7).$$

- Solve for $(14, 2)$

Question: Solving for $a + b + c$

The function $f(x)$ is defined by:

$$f(x) = \frac{x^2 + ax + b}{2x + c},$$

where a , b , and c are constants.

The graph of $f(x)$ does not intersect the line $x = 4$. If $f(5) = f(6) = 0$, what is the value of $a + b + c$?

- A. 10
- B. 11
- C. 12
- D. 13

Answer Explanation: Solving for $a + b + c$

Correct Answer: B.11

Solution:

- Since $f(x)$ does not intersect $x = 4$, the denominator must be 0 at $x = 4$:

$$2(4) + c = 0 \implies c = -8.$$

- Given $f(5) = 0$ and $f(6) = 0$, set the numerator to 0:

$$5^2 + 5a + b = 0, \quad 6^2 + 6a + b = 0.$$

- Solve:

$$25 + 5a + b = 0,$$

$$36 + 6a + b = 0.$$

- Subtract equations:

$$(36 + 6a + b) - (25 + 5a + b) = 0.$$

$$11 + a = 0 \implies a = -11.$$

Answer Explanation: Solving for $a + b + c$

- Find b :

$$25 + 5(-11) + b = 0 \implies b = 30.$$

- Compute sum:

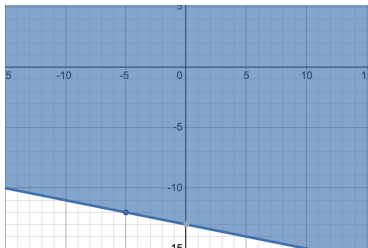
$$a + b + c = -11 + 30 + (-8) = 11.$$

Question: Finding $r + t$

The shaded region represents the solutions to the inequality:

$$rx + ty \geq -65$$

What is the value of $r + t$?



- A. 6
- B. 4
- C. -4
- D. -5

Answer Explanation: Finding $r + t$

Correct Answer: A.6

Solution:

- The boundary line of the shaded region is given by:

$$rx + ty = -65.$$

- The line passes through the points $(-5, -12)$ and $(0, -13)$.
- Substituting these into the equation:

$$r(-5) + t(-12) = -65, \quad r(0) + t(-13) = -65.$$

- Solving for r and t :

$$t = 5, \quad r = 1.$$

- Compute the sum:

$$r + t = 6.$$

Final Answer: 6.